

Storytelling

CMPSC 105 – Data Exploration



ALLEGHENY COLLEGE

Goals

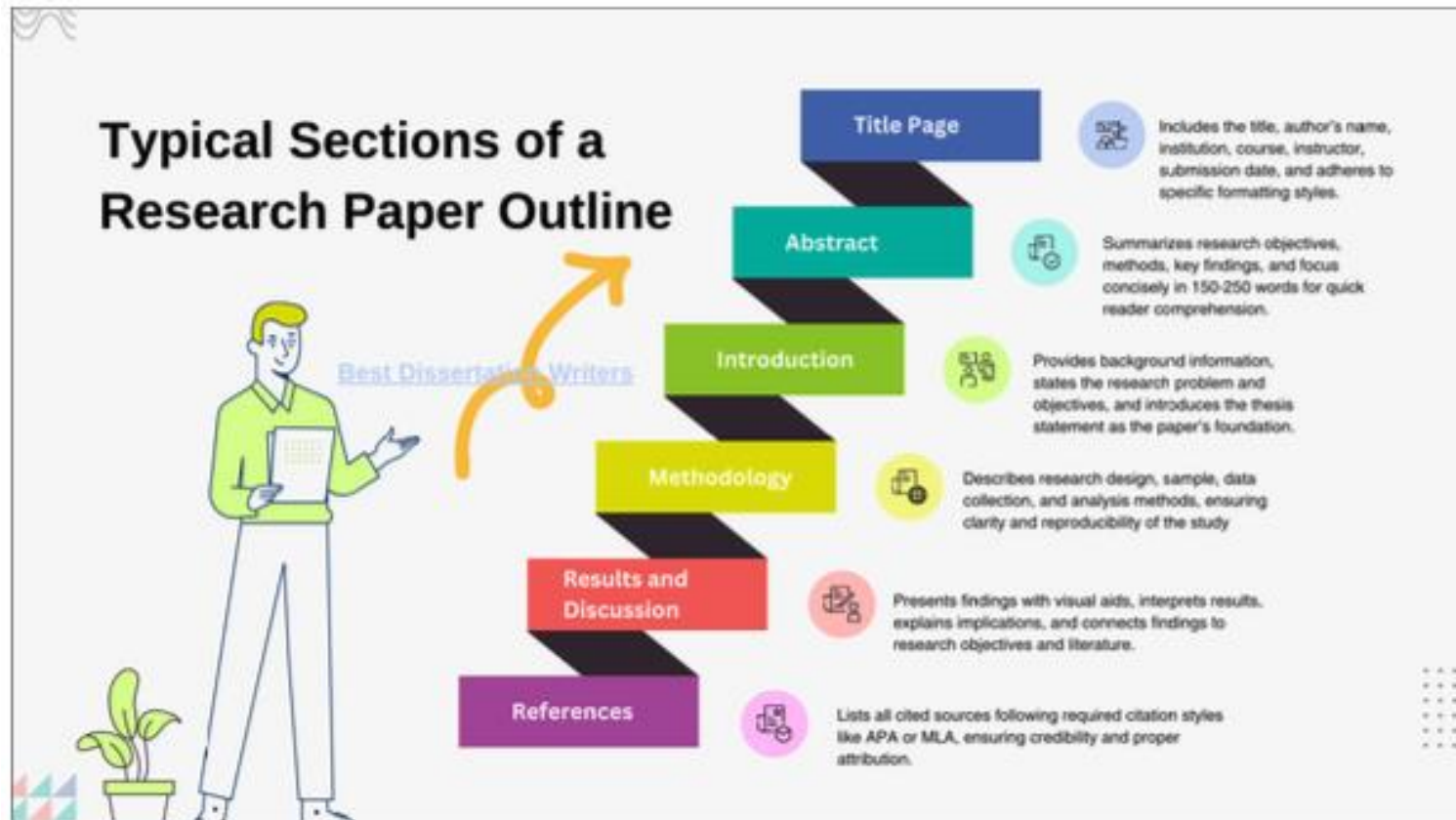
- Telling Your Story
- Supporting Your Story
- Finding Scholarly Research
- Tracing References
- Using References vs Explaining References

Telling Your Story

Main Steps

- Abstract
- Motivation
- Problem or Gap in Knowledge
- Hypothesis or Question
- Data
- Actionable Steps
- Evaluations
- Interpretation

Goals



Main Steps Continued

- Background
 - Motivation (more detail with ref)
 - Problem or Gap in Knowledge (more detail with ref)
 - Hypothesis or Question (more detail)
- Methods
 - Data
 - Actionable Steps
 - How evaluations will be done
- Results
 - Evaluations (with visualizations)
- Discussion
 - Interpretation of results (more detail)

Abstract

- Motivation
- Problem or Gap in Knowledge
- Hypothesis or Question
- Data
- Actionable Steps
- Evaluations
- Interpretation

Motivation Example

- Identify your research motivation
- **Do not** use personal goals
 - X Personal goal: “I wanted to learn how to do visualization because I think employers desire that skill”
- **Do** use personal or academic knowledge
 - Personal knowledge: “Individual golfers often have clubs made of different materials.”

Problem or Gap in Knowledge Example

- “It is unclear why having different materials matters.”

Hypothesis or Question Example

- “I want to know if the materials actually make a difference in how well the players play.”

Data Example

- “I will use information from public sources about which players use which materials and how highly ranked each player is.”

Actionable Steps Example

- “I will examine if players who use the same materials have similar rankings”

Evaluations Example

- “For each material, I will evaluate the “similarity” of rankings by computing the mean and standard deviation and comparing those to the mean and standard deviation for every other material”

Interpretation Examples

- Connect back to the hypothesis.

“If the rankings are found to not be the same from material to material, that indicates that materials may be related to player rankings. Players could therefore consider trying materials that highly ranked players use.”

“If the rankings are found to be the same from material to material, that indicates that materials may not be related to player rankings. Players could therefore try various materials.”

Full chain of logic

- **Motivation**, **Problem or Gap in Knowledge**, **Hypothesis or Question**, **Data**, **Actionable Steps**, **Evaluations**, **Interpretation**

Individual golfers often have clubs made of different materials. It is unclear why having different materials matters. This research is about understanding if the materials actually make a difference in how well the players play. Information from public sources about which players use which materials and how highly ranked each player is will be explored to answer the question. Specifically, players who use the same materials will be grouped together, and their rankings will be evaluated. For each material, the mean and standard deviation of the players rankings will be compared. If the rankings are found to not be the same from material to material, that indicates that materials may be related to player rankings. Players could therefore consider using the same materials that highly ranked players use.

Finding Scholarly Research

Finding Scholarly Research

- Where are you searching?
- What are searching for?
- Choosing useful papers out of the masses

Where?

- Databases!
- <https://allegheny.libguides.com/az.php>
- ACM (Association for Computing Machinery)
- ScienceDirect
- Taylor & Francis
- Scopus
- PubMed

What?

- Keywords!
- use keyword searches
- combine analysis method, field, topic, outcome measure
- example: classification, sports, golf, materials, player ranking

What? (classification, sports, golf, materials, player ranking)

Check all assumptions in your literature review

- Literature Review (LR) 1: GENERAL: does gear impact athletes?
 - search: sports relationship of player ranking with jersey material
 - search: sports relationship of player success with court preparations
 - search: sports relationship of player success with shoe weight
 - note: these searches probably will be about different sports, but that is good! It shows the relevance of your question broadly.

What? (classification, sports, golf, materials, player ranking)

- LR 2: NARROW SLIGHTLY: does golf club material impact golfers?
 - search: golf relationship of club material with hit accuracy
 - search: golf relationship of club material with hit strength
- LR 3: MOST NARROW: are the best golfers all using the same materials or is it something else?
 - search: classifying golfers by club materials
 - search: classifying golfers by physical attributes

What? (classification, sports, golf, materials, player ranking)

Check definitions even if you have an idea of what they mean.

- LR 4: DEFINE STANDARDS
 - search: golf player analysis (how are golfers usually evaluated?)
 - search: golf club materials analysis (what else has been examined and how?)
 - search: textbook chapter on classification (is the method applicable and how does it work?)
 - search: determinants of player ranking sports (where does ranking even come from?)

Choosing Papers

- Each search will probably bring up dozens of results
- Look for papers that have familiar authors
- Look for papers that are from reputable institutions
- Look for papers that have many citations
- For each search, try to find a foundational paper, and one on new developments

Tracing References

Find the foundation

- look at references list
- find oldest date with on-topic content
- repeat this from a few starting points

Find popular theories, stable

- look at references list
- what names appear in multiple articles?
- read intros
- what key ideas come up in multiple papers?

Find the latest, in development

- look at cited by list
- these are references that are more recent
- follow the links to newer papers

Why are you reading?

- You may read and reread the same set of papers for different reasons
- what analysis method did they use?
- What data did they use?
- What assumptions did they make?
- What was their take home message?

Using References vs Explaining References

Using References

- In a research narrative, use references to back up your points. Answer, “how do you know” for everything!

Individual golfers often have clubs made of different materials [References about variety of clubs player have]. It is unclear why having different materials matters [Ref questioning difference and similarities, perhaps competing views]. This research is about understanding if the materials actually make a difference in how well the players play. Information from public sources about which players use which materials and how highly ranked each player is will be explored to answer the question. Specifically, players who use the same materials will be grouped together, and their rankings will be evaluated. For each material, the mean and standard deviation of the players rankings will be compared. If the rankings are found to not be the same from material to material, that indicates that materials may be related to player rankings. Players could therefore consider using the same materials that highly ranked players use.

- Notice that the references did not have to be explained.